Neural estimation of a bivariate Hawkes process on high-frequency cryptocurrency data: a reproduction of Lee (2023) and a test of attention

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Abstract

We reproduce and extend the amortized neural estimation of a symmetric bivariate exponential Hawkes process proposed by Lee (2023), using mid-price changes of Binance BTCUSDT and ETHUSDT perpetual futures on a 100 ms grid. Under a frozen synthetic protocol (100,000 paths of 1,000 events, five seeds per network), the paper's LSTM attains a macro MSE of 0.0196 and a Transformer encoder 0.0237 ($p = 0.19$ across seeds): attention does not improve accuracy. The maximum likelihood estimator (MLE) has a macro MSE of 0.506, driven by the 1% of paths in which the decay rate is weakly identified; on a typical path it is as accurate as the networks and, with an analytic gradient, costs about 10 ms, the same order as a network at batch size one. On real windows, the Hawkes model beats a Poisson process in 99.8% of the cases but leaves serial dependence in the residuals, and its estimates lie outside the synthetic prior, with null cross-excitation. Fine-tuned from the synthetic weights on MLE labels of real windows, the networks reach a test macro MSE of 0.044 (LSTM, selected on validation) and 0.040 (Transformer), against 0.218 for the label mean. The Hawkes volatility implied by the MLE ranks windows like realized volatility (Spearman 0.78); the network's keeps the level but not the ranking (0.28). A pre-registered addendum, evaluated on a new week, shows that the selected LSTM was undertrained (a longer budget lowers its loss by 20%, from 0.026 to 0.021) and that averaging the LSTM and the Transformer is the most accurate estimator (0.020); it also shows that the 100 ms grid makes the MLE recover only about 58% of the continuous-time decay rate. Pretraining on a prior built from the real labels and simulated on the same grid cuts the loss on a second new week (March 2024) by 63% (0.109 to 0.040), the largest out-of-sample gain of this work. Trained on the true parameters of gridded paths, the LSTM undoes the grid bias in simulation, but the correction does not transfer safely to real windows.

Keywords: Hawkes processes, high-frequency data, cryptocurrencies, amortized inference, LSTM, Transformer
JEL classification: C13, C45, C58

1. Introduction

At short time scales a price is a sequence of discrete events, and these events cluster: a change tends to be followed by further changes, so that calm periods alternate with bursts.

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The homogeneous Poisson process cannot represent this dependence. The Hawkes process (Hawkes, 1971b,a), whose intensity rises after each event and then decays, can, and it has become a reference model of market microstructure (Bowsher, 2007; Bacry et al., 2013, 2015; Hawkes, 2018). Its branching ratio measures how much of the activity is endogenous (Filimonov and Sornette, 2012; Hardiman et al., 2013), and its parameters determine a model-implied volatility (Lee and Seo, 2017; Lee, 2024).

Estimation is usually done by maximum likelihood, which for exponential kernels is $O(n)$ per evaluation (Ozaki, 1979) and asymptotically efficient (Ogata, 1978), but requires an iterative optimization for every new window. Lee (2023) proposes to pay a training cost once instead: an LSTM is trained on simulated paths and maps an observed sequence directly to the four parameters of a symmetric bivariate Hawkes process of up- and down-moves. This is amortized, simulation-based inference (Cranmer et al., 2020), a strategy also used for spatial and extreme-value models (Gerber and Nychka, 2021; Lenzi et al., 2023) and, recently, for Hawkes processes observed only through counts (Lambe et al., 2026). A different line of work, the neural point processes (Du et al., 2016; Mei and Eisner, 2017; Zuo et al., 2020; Zhang et al., 2020; Shchur et al., 2021), uses networks to model the intensity itself; our goal is narrower, namely to estimate the interpretable parameters of a classical model.

Lee (2023) studied US equities with non-public NBBO data, left several protocol details unspecified, and considered a single compact recurrent architecture. Crypto-assets are a natural test bed: trading is continuous, best bid and offer data are public, and the market differs structurally from equities (Makarov and Schoar, 2020; He et al., 2022). Hawkes models have been fitted by MLE to jumps in one-minute crypto returns (Taddeo et al., 2024); here we work at the level of top-of-book events and with amortized neural estimation. We ask whether an attention-based architecture can estimate the parameters of a bivariate Hawkes process on cryptocurrency price changes with accuracy and stability equal to or better than Lee's LSTM, while keeping inference faster than the MLE. Three hypotheses were registered before the final run: (H1) Binance's public USD-M data yield sequences structurally equivalent to Lee's inputs; (H2) the LSTM attains error of the same order as the MLE at substantially lower latency; and (H3) attention improves error or robustness on sequences with long-range dependence.

Our contributions are the following.

- *A controlled reproduction.* We re-implement simulation, MLE and the LSTM, document the gaps in Lee (2023), and reproduce the paper's sampling distribution at its published parameter point.

- *A sharper reading of the synthetic comparison.* Per-parameter R^2, paired per-path tests and a diagnosis of the MLE show that the networks' advantage in aggregate MSE comes from a small set of weakly identified paths and from shrinkage toward the training prior, not from higher accuracy on a typical path; with an efficient MLE, the speed gain reported by Lee is not reproduced at batch size one.

- *A negative result for attention on synthetic data.* A Transformer encoder trained under the same protocol does not outperform the LSTM in any regime of the branching ratio.

- *A pre-registered empirical stage.* We characterize the mid-price events of BTCUSDT and ETHUSDT, separate the effect of the 100 ms grid from model misspecification

in goodness-of-fit tests, train networks on MLE labels of real windows (from scratch, fine-tuned and zero-shot) with a test block opened once, check robustness on a different month, and compare the implied Hawkes volatility with realized volatility.

Section 2 presents the model and the estimators, Section 3 the data and Section 4 the experimental design. Section 5 reports the results, Section 6 discusses them and Section 7 concludes.

2. Model and estimators

2.1. Symmetric bivariate Hawkes process

Let N_1 and N_2 count the up- and down-moves of the mid-price, with event times t_k^1 and t_k^2. The symmetric exponential model of Lee (2023) has intensities

$$\lambda_i(t) = \mu + \sum_{t_k^i < t} \alpha_{\mathrm{s}} e^{-\beta(t-t_k^i)} + \sum_{t_k^j < t} \alpha_{\mathrm{c}} e^{-\beta(t-t_k^j)}, \qquad i \neq j \in \{1,2\}, \tag{1}$$

with baseline μ, self-excitation α_{s}, cross-excitation α_{c} and decay β. Dominant cross-excitation describes short-term reversion (an up-move followed by a down-move) and dominant self-excitation describes persistence of direction (Bacry et al., 2013). The kernel mass matrix has largest eigenvalue

$$g = \frac{\alpha_{\mathrm{s}} + \alpha_{\mathrm{c}}}{\beta}, \tag{2}$$

the branching ratio; $g < 1$ is required for stationarity, and cascades lengthen as $g \to 1$ (Hardiman et al., 2013; Filimonov and Sornette, 2012). The stationary rate of each direction is $\Lambda = \mu/(1-g)$.

2.2. Maximum likelihood

For $\theta = (\mu, \alpha_{\mathrm{s}}, \alpha_{\mathrm{c}}, \beta)$ and horizon T, the log-likelihood is

$$\ell(\theta) = \sum_k \log \lambda_{c_k}(t_k) - \sum_{i=1}^{2} \int_0^T \lambda_i(u)\, \mathrm{d}u, \tag{3}$$

where c_k is the type of the k-th event. With exponential kernels the sums over the past obey a first-order recursion, so ℓ and its gradient cost $O(n)$ (Ozaki, 1979). We parametrize $(\mu, \alpha_{\mathrm{s}}, \alpha_{\mathrm{c}}, d) = \mathrm{softplus}(z)$ and set $\beta = \alpha_{\mathrm{s}} + \alpha_{\mathrm{c}} + d$, which enforces $g < 1$, and maximize with L-BFGS-B from five starting points: one based on the empirical event rate and four random perturbations in the unconstrained space. The MLE never receives the true parameter. The gradient is analytic and the recursions are solved as bidiagonal linear systems.

Consistency and efficiency are asymptotic. With 1,000 events and weak excitation, the kernel barely moves the intensity and the likelihood is nearly flat in β; as $\alpha_{\mathrm{s}} + \alpha_{\mathrm{c}} \to 0$, β is not identified. Near criticality, the mean rate $2\mu/(1-g)$ is well determined but μ and g separately are not. Both effects matter below.

2.3. Neural estimators

Each event is encoded as $(\log(1+\Delta t),\ s)$, with Δt the inter-arrival time and $s = \pm 1$ the direction. Both networks share the output head: a softplus produces $\mu, \alpha_{\mathrm{s}}, \alpha_{\mathrm{c}}, d > 0$ and $\beta = \alpha_{\mathrm{s}} + \alpha_{\mathrm{c}} + d$, so every estimate satisfies $g < 1$, as in Lee's robust variant. The loss is the MSE on the original parameter scale.

The *LSTM* follows Lee (2023): two LSTM layers of 12 units (Hochreiter and Schmidhuber, 1997) followed by a dense layer, 2,068 trainable parameters. The *Transformer* is an encoder (Vaswani et al., 2017) with a linear projection from 2 to 32 dimensions, sinusoidal positional encoding, two post-LN layers with four heads of dimension eight, a feed-forward block $32 \to 128 \to 32$, dropout 0.1 and temporal mean pooling, for 25,700 trainable parameters. Its attention cost is quadratic in the sequence length, and the post-LN variant usually benefits from learning-rate warm-up (Xiong et al., 2020), which we did not use.

Training on pairs (θ_i, x_i) with $\theta_i \sim \pi$ and squared loss minimizes the Bayes risk under the prior π, whose minimizer is the posterior mean $\mathbb{E}_\pi[\theta \mid x]$ (Lehmann and Casella, 1998; Sainsbury-Dale et al., 2024). On a test set drawn from π, no estimator has lower average MSE than the posterior mean, so beating the MLE there is expected and is not evidence of frequentist superiority. Also, the network does not extrapolate: its estimates shrink toward the high-density region of π, so data outside the support of the prior produce biased estimates.

2.4. Hawkes volatility

Let J be the absolute log return of an event, assumed i.i.d., and let $a_s = \alpha_{\mathrm{s}}/\beta$ and $a_c = \alpha_{\mathrm{c}}/\beta$. For the symmetric model the long-run variance of the log price per unit time is (Bacry et al., 2013)

$$\sigma_H^2 = 2\Lambda \left[\mathrm{Var}(J) + \frac{\mathbb{E}[J]^2}{(1 - a_s + a_c)^2} \right], \qquad \Lambda = \frac{\mu}{1 - a_s - a_c}. \tag{4}$$

The factor $(1 - a_s + a_c)^{-2}$ makes persistence ($a_s > a_c$) amplify variance and reversion damp it. We compare σ_H, scaled by the square root of the window duration, with the realized volatility of the log mid-price (Andersen et al., 2003; Barndorff-Nielsen and Shephard, 2002).

3. Data

The best bid and ask are needed to reconstruct the mid-price, and the official Binance archive provides a historical `bookTicker` stream for USD-M futures but not for spot.[1] We therefore use the BTCUSDT and ETHUSDT perpetual futures. Each downloaded file is recorded with its SHA-256, and audits check for positive prices, bid not above ask, time ordering and duplicate timestamps. Following Lee (2023), updates are reduced to a 100 ms grid keeping the last state of each cell, and an event is a change of the mid-price; the event type is its direction and the mark is its size in ticks.

The primary window is 15–28 January 2024; all 56 expected files (14 days, two symbols, `bookTicker` and `aggTrades`) were present. It is cut into non-overlapping windows of 1,000 events and split chronologically by day into calibration (15–22 January, 1,063 windows), validation (23–25 January, 476) and test (26–28 January, 274). The week of 23–29 October 2023 is a robustness window, used neither for training nor for selection (Table 1).

[1] Binance Public Data, `https://data.binance.vision`.

Table 1: Data coverage and characteristics of the mid-price events. Panel B uses the calibration block (15–22 January 2024).

Panel A: coverage		
	Primary (Jan. 2024)	Robustness (Oct. 2023)
Symbol-days	28	14
Events, BTCUSDT	887,803	652,773
Events, ETHUSDT	939,101	635,579
Events, total	1,826,904	1,288,352
Windows of 1,000 events	1,813	1,281
Panel B: event characteristics		
	BTCUSDT	ETHUSDT
Events	529,278	540,164
Fraction of up-moves	49.4%	49.5%
Repeats the previous direction	76.0%	75.7%
Median inter-arrival time	0.3 s	0.3 s
Inter-arrivals equal to the grid step	39.5%	37.3%
Median / 90th percentile jump (ticks)	15.5 / 48	10 / 29
One-tick jumps	9.7%	12.6%

Three features set these data apart from Lee's equities. About 76% of the changes repeat the previous direction, which points to persistence rather than reversion. The median jump is 10 to 15 ticks and only 10–13% are one tick, so at this scale the contracts behave as small-tick assets. And 37–39% of the inter-arrival times equal the grid step: the sampling discretizes time, whereas the model and the simulations are in continuous time.

4. Experimental design

Every decision is labelled as published information (R1), a reconstruction of a gap in the paper (R2), an adaptation to crypto (A1) or an extension (E1, the Transformer). The protocols below were fixed in decision records before the corresponding test sets were read.

4.1. Synthetic protocol

Paths are simulated by thinning (Ogata, 1981) from an empty history, the same assumption as in the likelihood, and have exactly 1,000 events. The frozen experiment has 75,000 training, 10,000 validation and 15,000 test paths. Parameters are drawn with $\mu \in [0.2, 0.6]$, $\beta \in [0.8, 2.0]$, self-excited fraction $\alpha_s/(\alpha_s + \alpha_c) \in [0.2, 0.8]$, and g balanced across three strata: low $[0.10, 0.35)$, intermediate $[0.35, 0.70)$ and near-critical $[0.70, 0.95]$. Lee does not publish his ranges, so these are R2. Each network is trained with five seeds, Adam with learning rate 10^{-3}, at most 300 epochs and early stopping with patience 30 on the validation loss (Table 2). There was no hyperparameter search for either network.

We report per-parameter and macro MSE, median absolute error, the MSE of g, validity and MLE convergence. Because the parameters have different scales, we also report the R^2 of each parameter relative to the prior mean. Networks are compared in two ways: across seeds (Welch and Mann–Whitney tests on five seeds), which measures training variability (Bouthillier et al., 2021), and across test paths (paired difference of the ensembles with a bootstrap interval and a Wilcoxon test), which measures the accuracy of the trained

Table 2: Synthetic training configuration and cost (five seeds per network, Apple Silicon with MPS).

	LSTM (R1/R2)	Transformer (E1)
Trainable parameters	2,068	25,700
Effective batch	64	8 (4 with accumulation)
Best epoch (per seed)	165, 164, 193, 239, 299	73, 42, 38, 60, 44
Training time per seed	5.4–14.8 h	16.6–28.8 h
Total time (5 seeds)	41 h	110 h

models (Efron and Tibshirani, 1993; Diebold and Mariano, 1995). The per-path analysis was added after the aggregate results were known and is therefore post hoc; it uses the frozen checkpoints, which reproduce the reported MSE to a relative difference of 10^{-8}, and changes nothing in the protocol.

Fixed point. To compare with the sampling-distribution table of Lee (2023), 2,000 paths are simulated at the published point $\mu = 0.3$, $\alpha_s = 0.4$, $\alpha_c = 0.7$, $\beta = 1.5$, which lies inside our prior. The networks are applied without retraining, and the MLE is run both with our heuristic starts and with Lee's initialization at the true value.

4.2. Empirical protocol

On real data there is no true parameter, so the empirical stage asks three narrower questions, fixed before the test block was opened: does the Hawkes model describe the windows, can a network reproduce the MLE on real windows, and does the estimate imply a volatility consistent with the realized one?

Goodness of fit. With the MLE of each calibration window, the time-rescaled residuals of each dimension are tested against Exp(1) by Kolmogorov–Smirnov (KS) and for serial dependence by Ljung–Box at 10 lags (Ogata, 1988; Brown et al., 2002), and the Hawkes model is compared with a homogeneous bivariate Poisson process by likelihood ratio and AIC. Two controls separate the grid from misspecification: real times jittered uniformly inside their 100 ms cell, with the MLE refitted; and paths simulated from each fitted model, kept in continuous time or mapped to the grid (one event per cell) and jittered in the same way.

Networks on real windows (scope A). Each window is labelled with its MLE. Both architectures are trained on the calibration labels with the loss, batch, epoch limit and patience of the synthetic protocol, in two variants, each with the five protocol seeds: from scratch (learning rate 10^{-3}) and fine-tuned from the synthetic weights of the same seed (learning rate 10^{-4}). Two references need no training: the synthetic networks as they are (zero-shot) and the mean of the training labels, the best constant predictor. The selected variant is the one whose five-seed ensemble has the lowest validation macro MSE. The test block is labelled and evaluated once, after the selection, with the stored weights; all variants are reported. The October week is labelled and evaluated in the same way. The machine budget was capped at 30 h; training used 3.0 h.

Volatility. Equation (4) is evaluated per window with the MLE and with the selected network, and compared with the realized volatility sampled every $\Delta \in \{0.1, 1, 5, 10, 30\}$ s. The metrics, fixed before the test was opened, are the median ratio σ_H/σ_{RV} and the Spearman correlation across windows, with $\Delta = 10$ s as the main horizon.

Table 3: Sampling distribution at the fixed point of Lee (2023) ($\mu = 0.3$, $\alpha_s = 0.4$, $\alpha_c = 0.7$, $\beta = 1.5$): mean and standard deviation over 2,000 paths of 1,000 events. Lee's rows use 100,000 paths of 2,000 steps.

	μ		α_s		α_c		β	
Estimator	Mean	SD	Mean	SD	Mean	SD	Mean	SD
LSTM (5-seed ensemble)	0.3044	0.0365	0.4950	0.0536	0.6245	0.0608	1.5318	0.1436
Transformer (5-seed ensemble)	0.2957	0.0388	0.5280	0.0561	0.6055	0.0620	1.5308	0.1541
MLE, heuristic (5 starts)	0.3065	0.0370	0.3956	0.0728	0.7075	0.0859	1.5156	0.1642
MLE, started at the truth	0.3065	0.0370	0.3956	0.0728	0.7075	0.0859	1.5156	0.1642
Lee (2023), network	0.3151	0.0358	0.4429	0.0661	0.5733	0.0697	1.5736	0.1364
Lee (2023), MLE	0.3036	0.0314	0.3988	0.0500	0.7024	0.0608	1.5078	0.1145

Table 4: Frozen synthetic test set (15,000 paths of 1,000 events). Macro MSE overall (with 95% t interval across five seeds) and by regime of g; MSE per parameter. Network rows are means over seeds.

	Macro MSE		Macro MSE by regime of g			
Estimator	Mean	95% CI	Low	Interm.	Near-crit.	MSE of g
LSTM	0.0196	[0.0133, 0.0260]	0.0199	0.0156	0.0233	0.0018
Transformer	0.0237	[0.0191, 0.0283]	0.0222	0.0190	0.0299	0.0021
MLE	0.5062	–	1.4806	0.0168	0.0212	0.0023
	MSE by parameter: μ / α_s / α_c / β					
LSTM	0.0023 / 0.0162 / 0.0166 / 0.0434					
Transformer	0.0024 / 0.0209 / 0.0212 / 0.0503					
MLE	0.0101 / 0.0061 / 0.0062 / 2.0023					

5. Results

5.1. Reproduction at the published fixed point

Table 3 reproduces the qualitative pattern of Lee (2023). The MLE is nearly unbiased, with standard deviations about $\sqrt{2}$ times Lee's, as expected for paths of half the length. The networks shrink α_s and α_c toward the centre of the prior in exchange for lower variance, as in the paper. The two MLE versions produce identical estimates, so initialization at the truth does not explain the MLE's accuracy at this well-identified point ($g \approx 0.73$). The macro MSE is 0.0103 for the MLE and 0.0111 for the LSTM ensemble.

5.2. Synthetic comparison: LSTM, Transformer and MLE

Table 4 summarizes the frozen test set. All estimators produce valid parameters on every path and the MLE converges on all of them (median of 14 iterations). The LSTM's mean macro MSE is 17.2% lower than the Transformer's, but with five seeds each the difference is not significant (Welch $p = 0.19$; Mann–Whitney $p = 0.056$) and the confidence intervals overlap. Part of the seed dispersion comes from the LSTM, which showed two convergence levels across seeds, with one seed stopping at the 300-epoch limit while still improving.

The MLE's aggregate error is almost entirely β (98.9% of its sum of squared errors), and its macro MSE in the low-g regime is 88 times that of the intermediate regime. Outside the low regime the MLE is competitive: 0.0168 against 0.0156 (LSTM) and 0.0190

Table 5: Post hoc analysis of the synthetic test set. Panel A: R^2 relative to the prior mean, macro median absolute error (AE) and MSE of g (network ensembles). Panel B: paired per-path comparison of the mean standardized squared error; median loss of A and B, share of paths where A is better, and bootstrap 95% CI of the mean difference A−B (negative favours A).

Panel A: accuracy relative to the prior mean

Estimator	R^2_μ	$R^2_{\alpha_s}$	$R^2_{\alpha_c}$	R^2_β	Median AE	MSE of g
Prior mean	−0.01	0.00	−0.01	0.01	0.1908	0.0670
MLE	0.24	0.90	0.90	−15.64	0.0674	0.0023
LSTM (ensemble)	0.84	0.82	0.81	0.65	0.0680	0.0017
Transformer (ensemble)	0.84	0.74	0.73	0.61	0.0765	0.0019

Panel B: paired comparison per path

A × B	Regime	Med. A	Med. B	A better	95% CI (A−B)
LSTM × Transformer	all	0.151	0.187	70.2%	[−0.051, −0.047]
	low g	0.112	0.131	61.1%	[−0.021, −0.016]
	intermediate g	0.127	0.164	72.8%	[−0.045, −0.041]
	near-critical g	0.237	0.304	76.7%	[−0.089, −0.081]
LSTM × MLE	all	0.151	0.149	48.8%	[−9.394, −0.809]
	low g	0.112	0.199	63.5%	[−26.977, −2.225]
	intermediate g	0.127	0.100	40.1%	[−0.040, −0.009]
	near-critical g	0.237	0.183	42.7%	[−0.467, −0.277]
Transformer × MLE	all	0.187	0.149	43.5%	[−9.376, −0.782]
	low g	0.131	0.199	59.4%	[−27.686, −2.207]
	intermediate g	0.164	0.100	34.2%	[0.002, 0.035]
	near-critical g	0.304	0.183	36.7%	[−0.374, −0.195]

(Transformer) in the intermediate regime, and 0.0212 against 0.0233 and 0.0299 near criticality. It also has the lowest MSE of the three for α_s and α_c.

The post hoc per-path analysis (Table 5) explains the gap. First, 1% of the paths, almost all with low g, carry 94.4% of the MLE's squared error in β (Fig. 1). This is an identification failure, not a numerical one: on each of the 300 worst paths the log-likelihood at the estimate exceeds that at the true parameter (median gain 2.80), and refitting 100 of them with 20 starts barely changes the result. Second, on the typical path the MLE is as good as the networks: its median standardized loss equals the LSTM's overall (0.149 versus 0.151) and is lower in the intermediate and near-critical regimes, and its R^2 for α_s and α_c is 0.90, against 0.82 and 0.81 for the LSTM. Third, path by path, the LSTM ensemble beats the Transformer ensemble in every regime, including the near-critical one, where long memory should favour attention.

Paired intervals for network minus MLE are negative overall even though the medians are equal, because the mean is dominated by the few extreme MLE errors. Panel A makes the same point: the networks never leave $[0.8, 2.0]$ for β, which bounds their worst case, while the MLE's R^2_β is -15.64.

Computational cost. During the final run, the MLE was timed at 105 ms per path while competing with neural training for the machine. A controlled benchmark of the same code, which reproduces the 15,000 recorded estimates exactly, gives the values in Table 6. With an analytic gradient and a linear recursion, the MLE costs a few milliseconds, the same order as a network at batch size one; the network advantage of about 150 times reported by

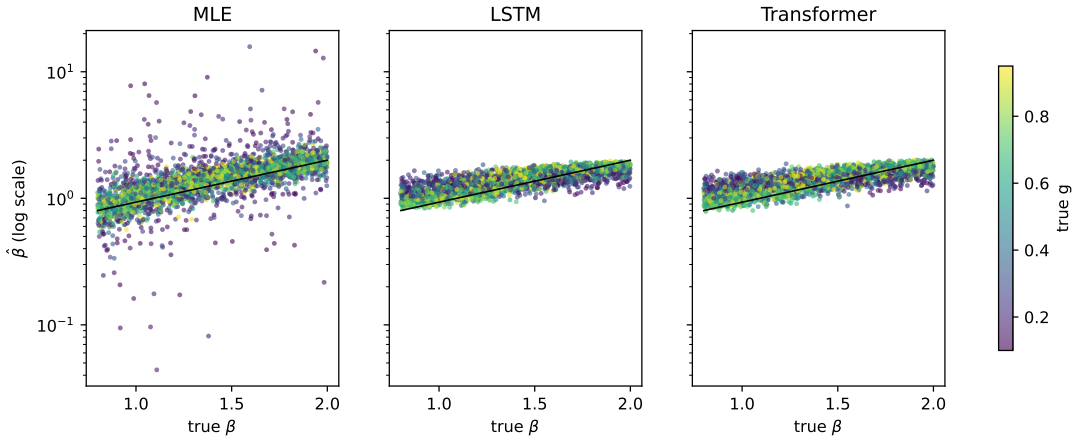


Figure 1: True versus estimated β on 3,000 synthetic test paths (logarithmic scale; colour is the true g). The networks shrink toward the centre of the prior; the MLE is unbiased over most of the domain but produces extreme values when g is low.

Table 6: Cost per path of 1,000 events in a controlled benchmark.

Estimator and execution mode	Time per path
MLE, 5 starts, one core	9.6 ms
MLE, 1 start, one core	2.2 ms
MLE, 5 starts, 4 processes (wall-clock)	2.5 ms
LSTM, batch size one, CPU (1 thread) / MPS	7.0 / 29.6 ms
Transformer, batch size one, CPU (1 / 10 threads) / MPS	9.9 / 4.8 / 5.0 ms
LSTM, batch of 256, MPS	0.07 ms
Transformer, batch of 32, MPS	3.7 ms
Lee (2023): network / MLE	12 / 1,763 ms

Lee (2023) is not reproduced. The networks' cost advantage appears in batched inference, above all for the LSTM. Lee does not describe his MLE. Fitting the same paths at his fixed point with implementations that differ in one factor at a time (Table 7) gives the same estimates in every variant and isolates the likelihood as the dominant cost: a double sum over pairs of events costs about 90 times as much as the recursion at 2,000 events, while a numerical gradient and Nelder–Mead add about a factor of five each. Lee's 1,763 ms falls between the two vectorized double-sum variants, so the reported ratio most likely reflects a likelihood computed without the recursion.

5.3. Real data: goodness of fit and the location of the estimates

The MLE converged on all 1,063 calibration windows. Table 8 and Fig. 2 summarize the residual diagnostics. The Hawkes model is far better than a homogeneous Poisson process: the likelihood ratio rejects Poisson at 1% in 99.8% of the windows, with a median AIC difference of -721. It is not the data-generating process, though. KS rejects in 70.3% of the jittered real windows, but the grid alone makes it reject in 47.7% of the simulated controls, so only part of that is misspecification. Ljung–Box keeps its nominal size on the simulated grid (4.7%) and rejects in 47.1% of the jittered real windows: part of the clustering remains in the residuals, beyond what a single exponential kernel absorbs. Jittering barely moves the estimates (median β from 2.71 to 2.77; median α_s from 1.34 to

Table 7: Cost of one MLE fit by implementation, at the fixed point of Lee (2023): median of 10 paths, one start at the true parameters, one core. The pure-Python double sum was timed for one evaluation and multiplied by the number of evaluations of the vectorized Nelder–Mead fit.

Likelihood	Gradient and optimizer	1,000 events	2,000 events
Recursion, vectorized	analytic, L-BFGS-B (this work)	1.1 ms	1.7 ms
Recursion, vectorized	numerical, L-BFGS-B	5.4 ms	8.3 ms
Recursion, Python loop	numerical, L-BFGS-B	47 ms	89 ms
Double sum, vectorized	numerical, L-BFGS-B	176 ms	711 ms
Double sum, vectorized	Nelder–Mead	818 ms	3,358 ms
Double sum, Python loop	Nelder–Mead (extrapolated)	26 s	105 s
Lee (2023), implementation not described			1,763 ms

Table 8: Residual diagnostics on the calibration block, per window and dimension. Under a correct model, about 5% of the tests reject at the 5% level.

Model and data	KS rejects	Median KS p	Ljung–Box rejects
Poisson, real data	100.0%	$< 10^{-47}$	40.6%
Hawkes, real data	100.0%	$< 10^{-8}$	46.2%
Hawkes, real data, jittered grid	70.3%	0.021	47.1%
Hawkes, simulated on the grid, jittered	47.7%	0.054	4.7%
Hawkes, simulated in continuous time	0.9%	0.626	4.9%

1.36). Repeated once on the 274 test windows, after the test was opened, the diagnostic shows the same pattern: Poisson is rejected in 100.0% of the windows (median AIC difference -905), and Ljung–Box rejects in 47.1% of the jittered real windows against 4.2% in the simulated grid controls.

The estimates fall far from the synthetic prior (Fig. 3). The median β is 2.71 (interquartile range 2.21–3.14), above the prior's upper limit of 2.0, and α_c is below 10^{-3} in 89.4% of the windows. Only 15.4% of the windows have β inside the prior, 0.4% have a self-excited fraction inside $[0.2, 0.8]$, and none satisfies all ranges at once. Null cross-excitation contrasts with Lee's equity fixed point, where $\alpha_c > \alpha_s$, and agrees with the directional persistence in Table 1. Since a squared-loss estimator approximates the posterior mean under its training prior, the synthetic networks cannot reach this region.

5.4. Neural estimators on real windows

Table 9 reports agreement with the MLE label. Applied as they are, the synthetic networks are worse than the constant label mean on validation (0.555 and 0.558 against 0.210), with strongly negative R^2 for α_s and β: they shrink into a prior that does not contain the data. Trained from scratch on 1,063 windows, both improve on the label mean in validation, but on the test block the LSTM is no better than it (0.250 against 0.218) and the Transformer beats it only modestly (0.169). Fine-tuned from the synthetic weights, both networks reproduce most of the label variance, and their test macro MSE is about five times lower than the label mean's. The fine-tuned LSTM was selected on validation (0.0359 against 0.0413 for the fine-tuned Transformer); on the test block the Transformer has the lower ensemble MSE (0.0396 against 0.0442), but a post hoc paired comparison per window gives an LSTM-minus-Transformer difference of 0.0046 with bootstrap 95%

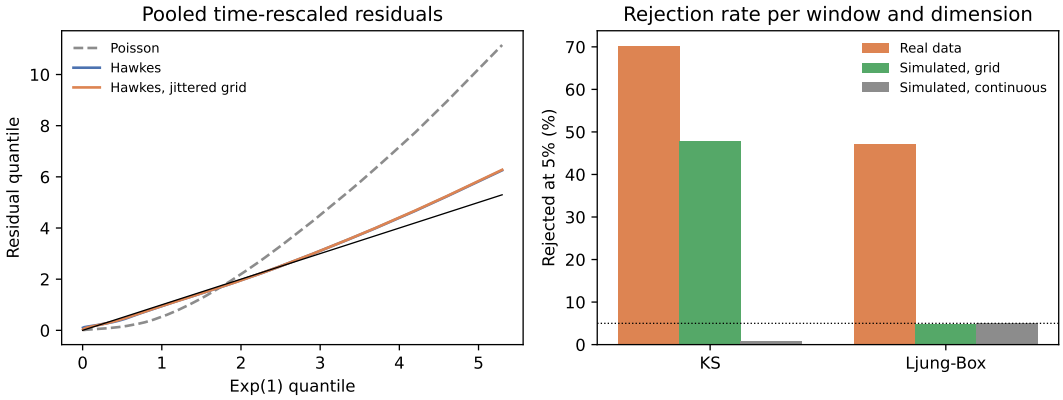


Figure 2: Left: Q–Q plot of the pooled time-rescaled residuals of the calibration windows against Exp(1). Right: rejection rates on real data and in the two simulated controls; the dotted line is the nominal 5%.

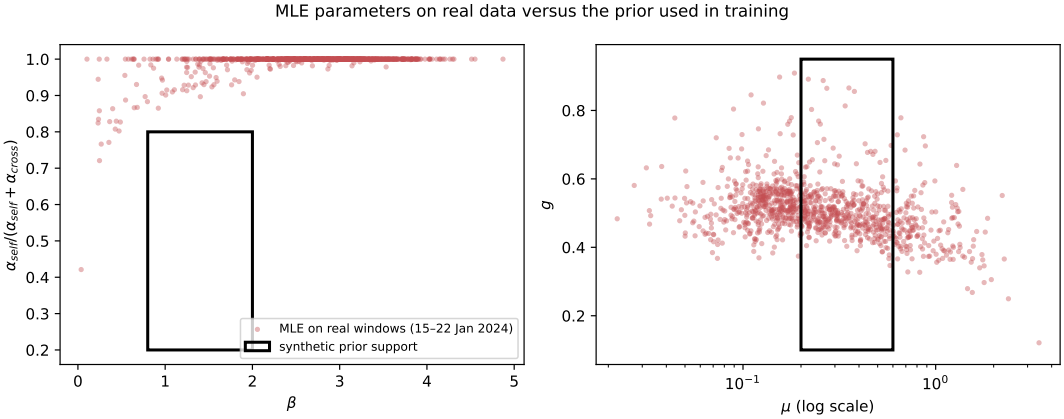


Figure 3: MLE estimates on the 1,063 calibration windows (points) and the support of the synthetic training prior (rectangles).

interval $[-0.0020, 0.0112]$ and Wilcoxon $p = 0.385$: the two cannot be told apart. The best epochs of the fine-tuned LSTM were 291–299 in every seed, close to the 300-epoch limit, so its training was cut by the budget.

5.5. Robustness: October 2023

The October week, never used for training or selection, yields 1,281 windows, and the MLE converged on all of them. Every variant has a higher loss than on the January test block (Table 10); the error of the label mean nearly triples (0.218 to 0.629), so the October labels lie farther from the January mean. The fine-tuned networks remain far ahead, and the zero-shot synthetic networks remain worse than the label mean. The fine-tuned Transformer now has a lower loss than the selected LSTM (0.083 against 0.121). In the same post hoc paired comparison, the LSTM-minus-Transformer difference is 0.0384 (bootstrap 95% interval $[0.0319, 0.0449]$, Wilcoxon $p < 10^{-32}$), and the Transformer has the lower loss in 64.3% of the windows. Under this change of regime, the fine-tuned Transformer degrades less.

Table 9: Agreement with the MLE label on real windows. Validation macro MSE of the five-seed ensemble and mean ± SD across seeds; R^2 per parameter on validation relative to the training-label mean; test macro MSE of the ensemble (block opened once). *Selected variant.

	Validation					Test
Variant	Ensemble	Seeds	R^2_μ	$R^2_{\alpha_s}$	R^2_β	Ensemble
Training-label mean	0.2096	–	0.00	0.00	0.00	0.2184
LSTM, synthetic weights	0.5550	0.5669 ± 0.0793	0.52	−4.48	−1.19	0.2501
Transformer, synthetic weights	0.5575	0.5707 ± 0.0161	0.59	−4.63	−1.08	0.2747
LSTM, from scratch	0.1385	0.1483 ± 0.0098	0.61	0.39	0.30	0.2502
Transformer, from scratch	0.1002	0.1060 ± 0.0012	0.93	0.50	0.47	0.1692
LSTM, fine-tuned*	0.0359	0.0454 ± 0.0133	0.90	0.85	0.82	0.0442
Transformer, fine-tuned	0.0413	0.0522 ± 0.0071	0.94	0.78	0.79	0.0396

Table 10: Agreement with the MLE label in the October 2023 robustness window: macro MSE of the seed ensemble, overall and by tercile of realized volatility. *Variant selected on the January validation block.

Variant	All	Low vol.	Medium vol.	High vol.
Training-label mean	0.6292	0.4997	0.3329	1.0550
LSTM, synthetic weights	1.1153	1.2838	1.2134	0.8487
Transformer, synthetic weights	1.0596	1.2621	1.1169	0.7997
LSTM, from scratch	0.4087	0.4052	0.3289	0.4920
Transformer, from scratch	0.3329	0.4486	0.3120	0.2383
LSTM, fine-tuned*	0.1212	0.1076	0.0712	0.1849
Transformer, fine-tuned	0.0828	0.0343	0.0386	0.1757

5.6. Hawkes volatility

Eq. (4) is finite in all windows, and its median absolute log change between consecutive windows (0.097) is close to that of realized volatility (0.102). Table 11 and Fig. 4 compare the two. With the MLE, the Hawkes volatility ranks the windows like the realized one (Spearman 0.78 on the test block at $\Delta = 10$ s) and is about 1.27 times higher at horizons of 5 to 30 s. The ratio falls from about 1.6 at 0.1 s to 1.27 at 10 s: realized variance per unit time grows with Δ, as expected when successive changes are positively autocorrelated, which is the persistence that self-excitation describes. Because Eq. (4) is a long-run quantity and the ratio is still about 1.3 at 30 s, either realized variance has not yet reached its long-run level or the model overstates persistence. With the selected network, the level is kept but the ranking is largely lost (Spearman 0.28 on the test block): the network shrinks its estimates toward the training-label mean, compressing the differences between windows.

5.7. Robustness of the attention hypothesis to a shift in β

The October result suggests a mechanism: the LSTM may store the decay in its weights, tuned to the January β, and degrade faster than the Transformer when β moves. We tested this in an exploratory addendum registered after the main results were frozen, which does not change them. Hawkes paths were simulated at the centre of the January labels ($\mu = 0.23$, $g = 0.5$, $\alpha_c = 0.001$), 400 per value of β, mapped to the 100 ms grid exactly as the real data and cut to 1,000 events. The target is each path's MLE, which is what the fine-tuned networks learned to imitate, and the frozen ensembles are used without

Table 11: Hawkes against realized volatility: median ratio σ_H/σ_{RV} and Spearman correlation ρ_S across windows, by sampling interval Δ. "Network" is the selected variant (fine-tuned LSTM ensemble).

		$\Delta = 1$ s		$\Delta = 10$ s		$\Delta = 30$ s	
Block	Parameters	Ratio	ρ_S	Ratio	ρ_S	Ratio	ρ_S
Calibration	MLE	1.37	0.87	1.27	0.74	1.30	0.59
	network	1.39	0.56	1.30	0.40	1.34	0.36
Validation	MLE	1.36	0.79	1.24	0.67	1.27	0.51
	network	1.37	0.59	1.24	0.47	1.28	0.37
Test	MLE	1.42	0.86	1.27	0.78	1.30	0.60
	network	1.56	0.34	1.38	0.28	1.45	0.33

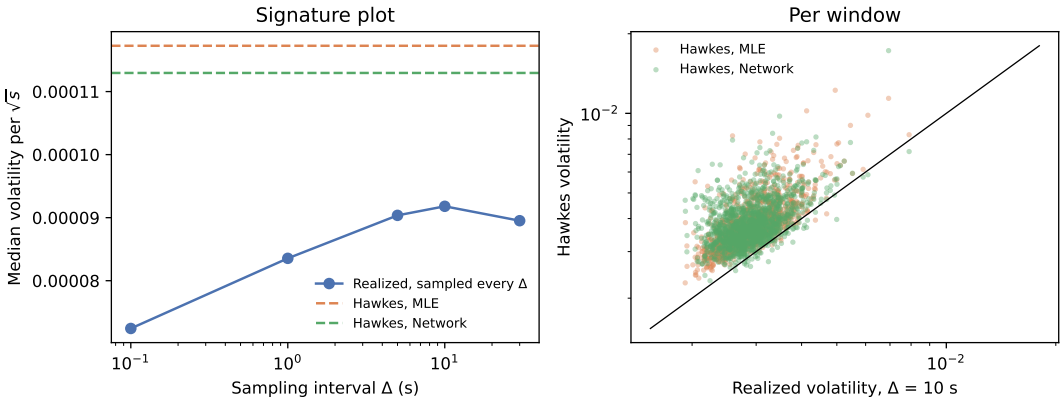


Figure 4: Calibration and validation windows. Left: median realized volatility per $\sqrt{s}$ against the sampling interval (signature plot) and the median Hawkes level. Right: Hawkes against realized volatility ($\Delta = 10$ s) per window.

retraining. The registered rule supports the hypothesis if the bootstrap 95% interval of the change in the mean LSTM-minus-Transformer loss between $\beta = 2.7$ and $\beta = 4.5$ lies above zero.

By the registered rule the hypothesis is supported: the difference grows by 0.067 from $\beta = 2.7$ to 4.5, with interval $[0.061, 0.073]$. Its premise, however, failed (Table 12). The grid merges events in the same cell and pulls the MLE β down, so a true β of 4.5 yields a median label of 2.73, the centre of January, not a shift away from it; on the grid the MLE recovers between 50% and 80% of the continuous-time β in this range. In the space of the labels, the LSTM has the lower loss inside and below the January range, where it is the Transformer that degrades; the two tie near the October median ($\beta \approx 3.2$); and above the training range (95% of the January labels lie below 3.74) the LSTM's loss grows about fourfold and crosses the Transformer's. The mechanism therefore holds only for β above the training range, and β alone does not explain the October gap, since October also has larger μ and α_s.

5.8. How much of the decay does the 100 ms grid hide?

The sweep shows that the grid pulls the MLE β down. To measure the effect over the region of the real labels, we simulated 100 continuous-time paths for each (μ, β) of a mesh (μ from 0.05 to 2, β from 1.2 to 13, $g = 0.5$, $\alpha_c = 0.001$), took them to the grid, cut them to 1,000 events and fitted the MLE. The median estimate defines a map from

Table 12: Mean loss against the MLE on simulated grid paths (ensemble of five seeds, fine-tuned weights). *Values added after the registered grid.

True β	1.2	2.0	2.7	3.5	4.5	6.0*	8.0*
Median MLE β on the grid	0.93	1.45	1.86	2.28	2.73	3.33	3.96
LSTM	0.017	0.007	0.022	0.028	0.028	0.037	0.125
Transformer	0.029	0.097	0.114	0.091	0.053	0.038	0.059

Table 13: Median MLE β on the 100 ms grid for continuous-time (μ, β), $g = 0.5$, 100 paths per cell.

$\mu \setminus$ true β	1.2	2.0	2.7	3.5	4.5	6.0	8.0	10.0	13.0
0.05	1.04	1.63	2.11	2.62	3.17	3.83	4.63	5.23	5.97
0.23	0.93	1.45	1.88	2.27	2.72	3.32	4.01	4.48	5.02
1.0	0.63	0.96	1.26	1.51	1.83	2.21	2.65	2.96	3.25
2.0	0.42	0.61	0.86	0.98	1.14	1.48	1.66	1.88	1.98

continuous-time parameters to grid labels, which we interpolate in logarithms and invert for each real window, a form of indirect inference (Table 13).

The bias grows with β and μ, that is, with the share of events that land in an occupied cell, and the branching ratio is also biased down (to between 0.22 and 0.48), while μ is nearly unaffected. Inverted, the median January label of 2.71 corresponds to a continuous-time β of 4.7 (5%–95%: 1.9–9.2; 95% of the windows inside the map): the grid label is a median 58% of the continuous value, and the mean memory of an event is about 210 ms rather than 370 ms. October's median of 3.19 corresponds to 6.5. Jittering the times inside their cell, as in the diagnostics, does not undo this bias, because it breaks ties but cannot restore merged events. The inversion assumes $g = 0.5$ and $\alpha_c \approx 0$ and gives the order of magnitude, not a corrected label per window.

5.9. A better fine-tuning recipe and a fresh test week

The fine-tuned LSTM reached its best epoch at 291–299 of 300 in every seed, and both fine-tuned networks predicted α_c worse than the label mean because β dominates the loss. Five new arms start from the same synthetic weights with the same data, seeds, learning rate and batch: a 1,000-epoch cap (LSTM); the cap with a five-epoch warm-up, halving of the learning rate after ten epochs without improvement and gradient clipping (both networks); and the latter with a training loss standardized by the variance of each label (both networks). A sixth candidate, registered before the new week was read, averages the predictions of the best LSTM and the best Transformer. One candidate is selected on the January validation block and compared, window by window, with the frozen fine-tuned LSTM on a week never used before: BTCUSDT and ETHUSDT, 5–11 February 2024, 711 windows labelled by the MLE and read once.

The longer cap was selected on validation (Table 14); its five seeds stopped at epochs 158 to 925. On the new week it beats the frozen LSTM by -0.0052 (paired bootstrap 95% interval $[-0.0060, -0.0044]$, Wilcoxon $p < 10^{-38}$), with the lower loss in 73% of the windows: the original LSTM was undertrained. Among the secondary comparisons (Holm-adjusted), the average of the two networks also beats the frozen LSTM (-0.0059, $p < 10^{-12}$) and has the lowest loss of all; the schedule ties; the standardized loss loses ($+0.005$ and $+0.007$), although it raises the R^2 of α_c from -21 to -1.0 (LSTM) and

Table 14: Macro MSE of five-seed ensembles on validation, on the new week and on October 2023 (exploratory).

Candidate	Validation	New week	October
LSTM, fine-tuned (frozen)	0.0359	0.0264	0.121
Transformer, fine-tuned (frozen)	0.0413	0.0285	0.083
LSTM, 1,000-epoch cap (selected)	**0.0294**	0.0212	0.116
LSTM, cap + schedule	0.0358	0.0260	0.131
Transformer, cap + schedule	0.0448	0.0295	0.094
LSTM, standardized loss	0.0354	0.0315	0.102
Transformer, standardized loss	0.0431	0.0330	0.085
Average of LSTM and Transformer	0.0304	**0.0204**	0.087
Mean of the training labels	—	0.1826	0.629

Table 15: Macro MSE of five-seed ensembles with the prior built from the real labels (scope B). Validation was recorded before March was read; February and October are exploratory.

Candidate	Validation	March	February	October
LSTM, fine-tuned (frozen)	0.0359	0.117	0.0264	0.121
LSTM, 1,000-epoch cap	0.0294	0.109	0.0212	0.116
Transformer, fine-tuned (frozen)	0.0413	0.112	0.0285	0.083
Average of the two above	0.0304	0.102	0.0204	0.087
Scope B LSTM, no fine-tuning	0.1439	0.122	0.2115	0.141
Scope B LSTM, fine-tuned	0.0202	**0.040**	0.0153	**0.034**
Scope B LSTM + Transformer	**0.0196**	0.056	**0.0137**	0.041
Mean of the training labels	—	0.582	0.1826	0.629

0.05 (Transformer), the only variants not clearly worse than the constant mean on that parameter. No Transformer arm beats the frozen LSTM on the new week, while in October the Transformers remain the best. Inside the training regime the LSTM is the better single network, and under a change of regime the Transformer is more robust. The average, whose two members err in different regions of β (Table 12), has the lowest loss on the new week and is close to the best in October.

5.10. A synthetic prior built from the real labels

The main protocol pretrained on a prior fixed in advance, outside the region of the real labels. A reduced version of the alternative, registered before any path was generated, builds the prior from the data. In 75% of the draws the label of a random training window is inverted to continuous time by the map of Section 5.8, with log-normal noise on μ and β and the branching ratio inflated by 1 to 1.25 to offset the grid; in 25% the parameters come from a wide box around the real range. Each of 22,000 paths is simulated in continuous time, taken to the 100 ms grid, cut to 1,000 events and labelled by the MLE on the grid, so that the synthetic target means the same as the real one; the real labels fall inside the 1–99% synthetic range in 98–99.8% of the training windows. The LSTM is pretrained on these paths (150-epoch cap, five seeds) and fine-tuned with the 1,000-epoch recipe above. Since February had been read, a second unseen week was read once: 4–10 March 2024, 1,860 windows.

Table 16: Error against the continuous-time parameters on 2,000 new simulated paths (median relative error of β: signed and absolute).

Estimator	Macro MSE	MSE β	Bias β	|Error| β
LSTM, trained on the truth	**0.328**	**0.90**	+2.8%	8.9%
MLE on the grid	3.951	11.68	−44.1%	44.1%
MLE corrected by the map	0.654	1.59	−2.4%	**8.4%**

The registered primary comparison, the fine-tuned scope B LSTM against the LSTM with the 1,000-epoch cap on March, is favourable: -0.068 (95% interval $[-0.082, -0.056]$, Wilcoxon $p < 10^{-10}$), a 63% lower loss, with the lower loss in 72% of the windows (Table 15). All three secondary comparisons also favour scope B after Holm's correction, and the sign is the same on February and October. The gain is concentrated where the earlier networks failed: March and October are the weeks farthest from calibration (the label mean errs three times more there than on February), and most of the error of the earlier LSTM is in μ (March MSE 0.31, $R^2 = 0.61$), which the scope B network brings to 0.086 ($R^2 = 0.89$); β also improves (0.097 to 0.064). Without fine-tuning the scope B network is no better than the earlier ones, so the level of the estimates comes from the fine-tuning. Unlike on February, averaging with the Transformer does not help on March or October.

5.11. Can a network undo the grid?

The MLE on the grid cannot recover the continuous-time β, but a network trained on gridded paths with the true parameters as target can learn the correction. With the protocol registered before training, the LSTM was trained from scratch on the scope B paths with their continuous-time parameters as target and compared on 2,000 new paths of the same prior with the MLE on the grid and the MLE corrected by inverting the map of Section 5.8.

The network beats the corrected MLE in the registered primary comparison: -0.326 (95% interval $[-0.405, -0.256]$), with the lower loss in 60% of the paths (Table 16). The map alone removes almost all of the bias and is the more precise on a typical path; the network wins through the tails of α_s and β, where the map, measured at a single branching ratio, extrapolates poorly. On real windows, however, the two corrections disagree: the network implies a median continuous-time β of 7.7 on validation and 6.3 on February, against 4.9 and 3.1 from the map (Spearman 0.55). The real windows are not Hawkes paths, so a correction learned in simulation does not transfer safely, and without the continuous-time truth neither can be checked.

5.12. Hawkes volatility with the addendum networks

Applied post hoc to February (711 windows), with the comparison registered before computing it, the Hawkes volatility of the LSTM+Transformer average ranks windows like realized volatility with Spearman 0.63, against 0.47 for the frozen LSTM (difference 0.155, bootstrap 95% interval $[0.130, 0.181]$); the Transformer alone reaches 0.69 and the MLE 0.77. Training the LSTM for longer barely helps (0.49). All estimators get the level right (median ratio 1.3–1.4). The LSTM has the lower parameter error, but the Transformer preserves more of the differences between windows, on which the ranking depends.

6. Discussion

6.1. The hypotheses

H1 is supported as to structure and volume. The public data yield sequences with the structure of Lee's inputs in sufficient volume. Their content differs from equities, however: persistent direction, multi-tick jumps and times discretized by the grid.

H2 is partially supported. The LSTM's error is of the same order as the MLE's (0.0111 against 0.0103 at the fixed point; practically equal median per-path loss on the synthetic test). The substantially lower latency is not confirmed at batch size one against an efficient MLE; it holds for batched inference.

H3 is not supported on synthetic data. The Transformer did not reduce error: its mean macro MSE was higher, without significance across seeds, and its ensemble lost to the LSTM's path by path in every regime, including the near-critical one. Its advantages were latency on MPS (5 ms against 30 ms) at the cost of 2.7 times longer training. On real windows, the fine-tuned architectures cannot be told apart on the test block, and in October the Transformer has the lower loss in 64% of the windows. This is the only evidence in favour of H3, and it comes from a post hoc comparison rather than the pre-registered selection. The β sweep (Section 5.7) does not explain it by a shift in β alone. On a pre-registered new week no Transformer arm beat the frozen LSTM, but the average of the two did (Section 5.9), so attention helped as a complement to the LSTM. That complement depends on the prior: once the LSTM is pretrained on a prior built from the real labels (Section 5.10), it is the best estimator on March and October by itself, and the average no longer helps. The Transformer keeps one advantage, for volatility: it preserves more of the ranking of windows (Section 5.12).

6.2. Why the networks win on aggregate MSE

The MLE's synthetic macro MSE is about 25 times the networks', but the gap has two sources, neither of which is greater accuracy on a typical path. The first is identifiability: with low g the MLE returns extreme values of β that legitimately maximize the likelihood. The second is the prior: the networks approximate the posterior mean under the training distribution and never leave its range, which bounds their worst case at the price of shrinkage bias. This robustness holds only if the prior represents the data, which is not the case on crypto. Lee's qualitative conclusions survive (error of the same order as the MLE, shrinkage, a nearly unbiased MLE at a well-identified point); absolute MSE values are not comparable across the two studies because they depend on parameter ranges that the paper does not publish. Two points diverge: the speed gain of two orders of magnitude, which a double-sum likelihood recovers (Table 7), and the use of aggregate MSE alone, which hides identifiability failures that initialization at the true value tends to mask.

6.3. What the real data show

Price changes in these contracts cluster strongly, but a single exponential kernel does not capture all of the clustering, and the grid explains part, not all, of the departure from Exp(1). On real data the target of the networks is therefore the MLE of an approximating model, observed through the grid, whose label is about 58% of the continuous-time decay rate (Section 5.8). Its parameters ($\alpha_c \approx 0$, $\beta \approx 2.7$) lie outside the synthetic prior, so synthetic training does not transfer directly; it is useful as an initialization, since only fine-tuned networks explain most of the label variance with 1,063 training windows. The

reduced scope B shows the effect of the prior: a prior built from the real labels and simulated on the grid cuts the out-of-sample loss by almost two thirds, mostly by extrapolating μ to activity levels the calibration week did not cover. For volatility, reproducing the label on average is not enough: the selected network keeps the level of the Hawkes volatility but loses most of the cross-sectional ranking that the MLE preserves; the Transformer recovers much of it, but not all.

6.4. Limitations and threats to validity

Construct. On real data the label is the MLE of a model that the residual tests reject in part, and α_c lies on the boundary of the parameter space, where the usual asymptotic theory does not hold; the labels are biased by the grid, which jittering does not undo and which a network trained on the truth corrects only in simulation. *Prior.* The synthetic metrics depend on ranges that were not calibrated on the data; the reduced scope B, which calibrates them, was run for the LSTM only and with a single prior design. *Statistical.* Five seeds capture only part of training variability; one synthetic LSTM seed and all fine-tuned LSTM seeds were limited by the 300-epoch budget, which the addendum confirms; the comparison between fine-tuned networks on the test and October blocks is post hoc. *Architecture.* Neither network was tuned, the batch sizes differ (64 against 8), and the post-LN Transformer was trained without warm-up. *Internal.* Leakage across windows and tuning on test data are mitigated by chronological splits and a test block locked behind an explicit flag and opened once. *External.* Two assets, one venue and two months do not represent the market; the empirical test block covers three days, including a weekend. *Computational.* Latency depends on hardware, batch size, warm-up and contention between processes.

7. Conclusion

We reproduced the qualitative findings of Lee (2023) at the published parameter point and compared the paper's LSTM with a Transformer encoder under a frozen protocol on public cryptocurrency data. On synthetic data, attention brought no gain in accuracy. The networks' advantage over the MLE in aggregate MSE comes from robustness on weakly identified paths and from shrinkage toward the prior; on a typical path the MLE is equally accurate and, when implemented efficiently, equally fast at batch size one. On BTCUSDT and ETHUSDT perpetual futures, the Hawkes model captures most but not all of the clustering, with parameters in a different region from equities and no cross-excitation. Networks reproduce the MLE on real windows only when fine-tuned from synthetic weights; fine-tuned, the LSTM and the Transformer are equivalent on the test block, and the Transformer degrades less in a different month. On a pre-registered new week, the LSTM trained for longer and the average of the two networks both beat the selected LSTM, the average by the widest margin; on a second new week, an LSTM pretrained on a prior built from the real labels cut the loss by almost two thirds. The answer to the research question is negative for accuracy on synthetic data and open for robustness on real data.

Natural next steps are to extend the prior built from the real labels to the Transformer and to more weeks and assets; to make the grid correction trustworthy on real data, with a grid-aware likelihood (Lambe et al., 2026) and a prior that covers the misspecification of the market; to replace the single exponential kernel, as the residual dependence suggests; to

tune both architectures without touching any test block and to confirm the October result on a pre-registered week outside the training regime; and to train directly for volatility with a loss that preserves the ranking of windows.

Data and code availability

The market data are public (Binance Public Data). The code, configuration files, decision records and generated tables are kept in the project repository; the numbers in this paper are taken from tables generated by script from the saved artifacts.

Appendix A. Reproduction ambiguities in Lee (2023)

The paper mentions paths of 2,000 steps but an input shape of 1,000; our protocol fixes 1,000 events and does not compare lengths. It does not publish the parameter ranges, seeds, batch size or the NBBO data. It calls the two LSTM layers followed by a dense layer "three layers"; our implementation follows that structure. The $\log(1 + \Delta t)$ input and the ± 1 direction encoding are our reconstructions (Lee uses inter-arrival times and types $1/2$). All such decisions are marked R2 in the project records.

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